Experiment 13.1.1

Algorithm –

1. Start
2. Input
3. Read max_value from the user.
4. Initialize
5. Create an empty list fib_sequence
6. Set a = 0 and b = 1
7. Loop Condition
8. While a ≤ max_value, repeat steps 5–6:
9. Store Value
10. Append a to fib_sequence
11. Update Values
12. Set a = b
13. Set b = a + b (using previous values)
14. Exit Loop
15. When a > max_value, stop the loop
16. Output
17. Return (or display) fib_sequence
18. End

Excecution –

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13.1.1. Fibonacci Sequence using Modules

01:46

Write a Python program using modular programming to generate the Fibonacci sequence up to a given maximum value.

Module Requirements

- Create a module **fibonacci_module.py** with a function **generate_fibonacci_sequence(max_value)** that returns list of all Fibonacci numbers less than or equal to the maximum value.

Main Program Requirements

Create **main.py** that performs the following:

- Import the **fibonacci_module.**
- Read an integer n from user input.
- Validate if n is positive.
- If valid, call the module function and display the Fibonacci sequence.
- If invalid, display an error message.

Input Format:

- Single integer n representing the maximum value for Fibonacci sequence.

Output Format:

- If $n > 0$, print the Fibonacci sequence as space-separated numbers.
- If $n \leq 0$, print: "Please enter a positive integer".

Note:

- Two files are provided: **fibonacci_module.py** and **main.py**. The **main.py** file handles input and output. You need to complete both files based on the question requirements.

Sample Test Cases +

Explorer

main.py fibonacci_... Submit

```python
import fibonacci_module

n = int(input())

if n > 0:
    fib_series = fibonacci_module.generate_fibonacci_sequence(n)
    print(' '.join(map(str, fib_series)))
else:
    print("Please enter a positive integer")
```

Debugger

Average time	Maximum time
0.007 s	0.014 s
7.38 ms	14.00 ms

4 out of 4 shown test case(s) passed

4 out of 4 hidden test case(s) passed

Test case 1 12 ms Debug

Expected output	Actual output
10	10
0 1 1 2 3 5 8	0 1 1 2 3 5 8

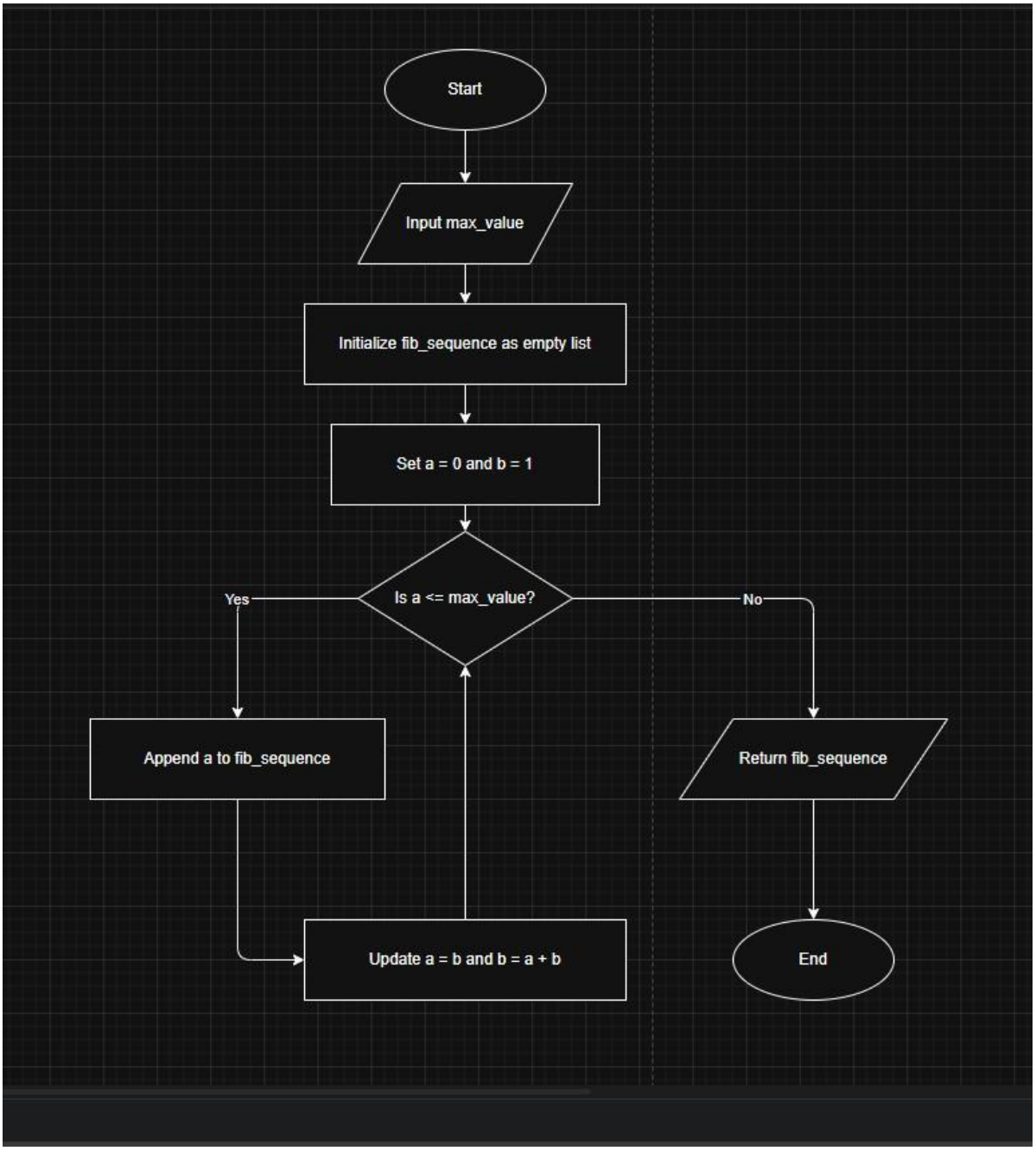
Test case 2 5 ms

Test case 3 4 ms

Terminal Test cases

‹ Prev Reset Submit Next ›

Flowchart –


Start
Input max_value
Initialize fib_sequence as empty list
Set a = 0 and b = 1
Is a <= max_value?
Yes
No
Append a to fib_sequence
Return fib_sequence
Update a = b and b = a + b
End